13.4 Apiculture

Ever since the beginning of civilization, man has been trying to make use of organisms around him for various purposes and to rear them for increasing their number. One of the finest discoveries is our knowledge regarding the procurement of honey collected by honey bees. Care and management of honey bees on a commercial scale for the production of honey is called **Apiculture** or **Bee Keeping**. The word 'apiculture' comes from the Latin word '*apis*' meaning bee. Bees are reared in apiaries that are areas where a lot of bee hives can be placed. There are five well recognized types of bees in the world. They are *Apis dorsata* (Rock bee), *Apis florea* (Little bee), *Apis indica* (Indian bee), *Apis mellifera* (European bee) and *Apis adamsoni* (African bee).

Social organization of honey bees

In honey bees, a highly organized division of labour is found. A well developed honey bee colony consists of the Queen, Drones and Workers (Figure 13.4). All the three types depend on each other for their existence. There is normally one queen, 10,000 to 30,000 workers and few hundred drones (male bees) in a colony.

Queen bee is a functional female bee present in each hive and feeds on Royal Jelly. Its sole function is to lay eggs throughout its life span. The virgin queen bee mates only once in her life. During the breeding season in winter, a unique flight takes place by the queen bee followed by several drones. This flight is called ***"nuptial flight".*** The queen bee produces a hormonal chemical substance called pheromone. The drones in the area are attracted to the pheromone and mating takes place. During mating, the drone releases large number of sperms for sufficient fertilization. In a life span of two to four years, a queen bee lays about 15 lakh eggs. When the queen bee loses its capacity to lay eggs, another worker bee starts feeding on the Royal Jelly and develops into a new queen.

Among the honey bees, **workers** are sterile females and smallest but yet function as the main spring of the complicated machinery in the colony. Worker bee lives in a chamber called 'Worker Cell' and it takes about 21 days to develop from the egg to adult and its lifespan is about six weeks. Each worker has to perform different types of work in her life time. During the first half of her life, she becomes a nurse bee attending to indoor duties such as secretion of royal jelly, prepares bee-bread to feed the larvae, feeds the queen, takes care of the queen and drones, secretes bees wax, builds combs, cleans and fans the bee hive.



Honey bee uses its long-tube like tongue to extract sugary liquid called nectar from the flowers. The nectar is stored in the stomach and the enzyme *invertase* transforms it into honey. This increases the storage life of honey and also contributes to its medicinal value.

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